

Core Question

The model requires a ground-truth-derived teleportation matrix E_i as input in order to simulate a given time frame. How, then, can it *forecast* a future time frame for which no ground truth exists?

How the Model Works

We process one month of urban traffic data, divided into 15-minute frames. For each frame i the model computes:

$$G_i = F(T, E_i),$$

where the two inputs are:

- T : the **transition matrix**, built from network topology and link features.
This is independent of ground truth and is always available.
- E_i : the **teleportation matrix**, derived *directly from the ground-truth traffic distribution* for frame i .
This is the problematic input: it encodes the very answer the model is meant to produce.

What We Observe

Observation 1 — Same-frame evaluation

When evaluated on the *same* time frame that produced E_i :

$$E_i \approx G_i \quad (\text{relative error} \approx 0.02).$$

This is unsurprising. The identity map $I(E_i) = E_i$ achieves the same result trivially. The model ingests E_i , applies stochastic processing, and returns an output nearly identical to the input, rendering the computation **redundant**.

Observation 2 — Cross-week prediction

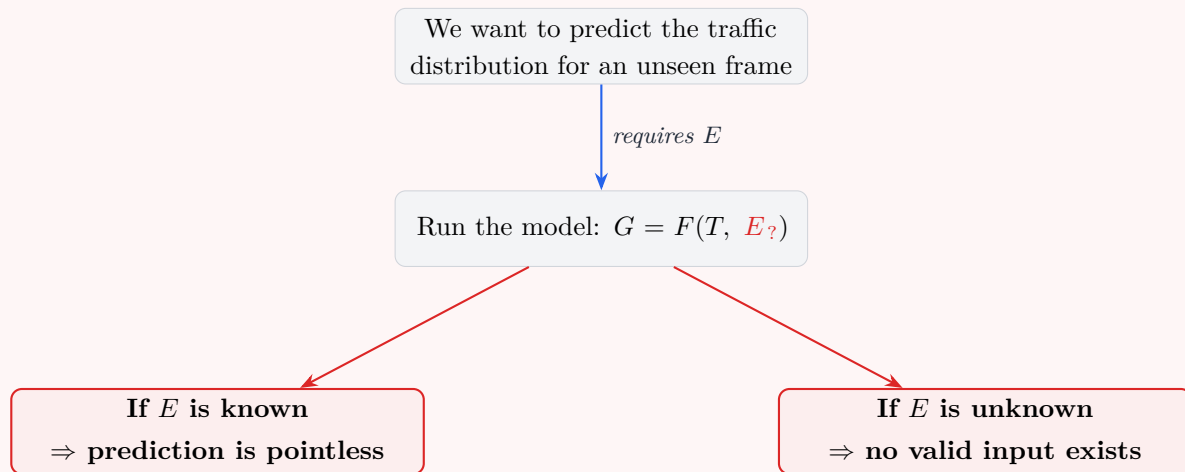
To test genuine predictive power, we fed the teleportation matrix from a *Week 1* frame (E_{i_1}) into F and measured how well it predicted the corresponding *Week 2* frame (E_{i_2}):

$$\frac{\|G_{i_2} - E_{i_2}\|}{\|E_{i_2}\|} \approx 0.4.$$

Although this is a large improvement over earlier models (which had relative errors ~ 300), a **40% relative error is still too high for reliable forecasting**.

The Circular-Dependency Problem

The architecture contains a fundamental circular dependency. Consider the real use-case: *simulate traffic in Salt Lake City for a specific 15-minute window in March 2026.*



In both branches, the model fails to serve as a practical simulator:

1. **Ground truth is available.** Feeding it in is redundant, we already have what we want.
2. **Ground truth is unavailable** (the realistic case). There is no valid teleportation matrix to supply, so the model *cannot run*.

The Open Question

What should be used as the teleportation input E so that the model becomes a viable, practical simulator for *unknown* future time frames?